

Persistent Latent Trajectory Features for Sequential Forecasting

Anonymous Author(s)

Abstract

Sequential data such as videos and time series are frequently modeled through latent representations, but these representations can miss higher-order structure in the evolving signal. In this work, we study whether topological summaries of latent trajectories can improve latent-space predictions. Given a sequence, we embed each state into a latent vector, form point clouds, compute persistence features, then combine them with the latent state as input to a downstream predictor. Across 9 video and time series datasets for 3 encoder families, adding topological features improves prediction over latent-only baselines in 24/27 settings and outperforms PCA, KPCA, manifold, and random-feature controls. Our experiments show that these gains are not explained solely by additional dimensions or generic linear, kernel, manifold, or nonlinear features, but instead arise from complementary predictive structure captured by persistent topology.

CCS Concepts

• Computing methodologies;

Keywords

machine learning, topology, autoencoder, representation learning

1 Introduction

In videos or dynamic systems, we often want to predict future observations, which is a computationally intensive task. Instead, many approaches operate on compressed latent representations learned by models such as autoencoders or convolutional neural networks (CNNs) [25, 32, 33]. However, these states might discard structural information that is useful for forecasting, especially when the sequence contains connected components, loops, or recurring geometric patterns. This motivates augmenting learned latent trajectories with topological descriptors computed from recent latent states. Although we focus on next-latent forecasting, the proposed representation is applicable to downstream tasks such as classification and anomaly detection [4, 22].

Existing latent forecasting methods typically predict future states from current or recent latent representations [7]. However, it remains unclear whether the topology of a latent trajectory contains complementary predictive information, or whether any improvement can instead be explained by generic nonlinear expansions or manifold geometry. We tackle this by computing persistent homology over sliding windows of latent states, rather than over individual frames, and compare it against linear (PCA), nonlinear (KPCA and random Fourier features), and manifold-learning (Laplacian Eigenmaps and Diffusion Maps) controls.

In this paper, we study whether persistent topology descriptors improve next-latent forecasting for sequential data. We encode each sequence state into a latent vector, compute Vietoris–Rips persistence on sliding windows of recent latent states, vectorize the resulting persistence diagrams, and use these descriptors as auxiliary inputs to an LSTM predictor. We evaluate 9 datasets across standard,

geometry-regularized, and topology-regularized autoencoders. Perturbation and recoverability tests examine whether topology provides stable and complementary information beyond the latent representations, while PCA, KPCA, Laplacian eigenmaps, diffusion maps, and random Fourier feature controls test whether improvements arise from persistence topology rather than generic geometric or nonlinear representations. Overall, topology provides the largest gains when combined with geometry-preserving latent representations, implying that preserving latent geometry is a condition for effective topological augmentation.

1.1 Background: TDA

Topological Data Analysis (TDA) is a framework that uses algebraic topology to analyze data and extract topological descriptors [13], with persistent homology as its main technique [23, 38]. Persistent homology tracks the birth and death of topological features across a filtration [23], producing summaries such as persistence barcodes or persistence diagrams [14]. Figure 1 illustrates the Vietoris–Rips (VR) filtration process. These summaries can then be vectorized and used in statistical analysis or machine learning [55].

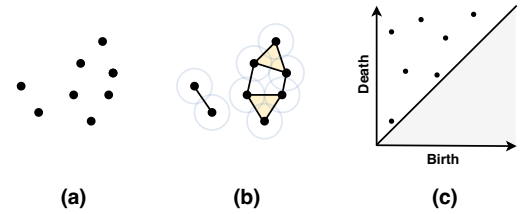


Figure 1: The Vietoris–Rips persistence. (a) Input point cloud. (b) Growing balls around each point define the Vietoris–Rips filtration: overlapping pairs include edges, and higher-order overlaps induce filled simplices. (c) The resulting persistence diagram records when topological features appear and disappear across the filtration.

The main homology dimensions used in this paper are H_0 , which describes connected components, and H_1 , which describes loop-like structures. We also use many vectorizations of persistence diagrams such as Betti curves, persistence images, persistence landscapes, and PersLay-style features. Betti curves count the number of active topological features across filtration values, while persistence images and landscapes retain more information about birth-death locations in the persistence diagram [23].

Persistence-based methods are used in many fields, including material science to detect cavities [31], hurricane analysis [49], and cosmology to describe Cosmic Web structures [39]. Persistence homology is also robust under small input noise [19], making it useful for applications such as denoising [30], anomaly detection

[54], image segmentation [18], and feature extraction for machine learning [9].

TDA is applicable to many modalities, such as time series [28], videos [48], images [1], point clouds [13], and graphs [2]. For time series, a common approach is to construct a point cloud using time-delay embedding [45], then applying a distance-based filtration such as VR [24]. In contrast, our main pipeline constructs point clouds from sliding windows of learned latent states and computes VR persistence on them, avoiding a separate delay-embedding step.

1.2 Related Work

Persistent homology is extensively used as a feature representation, regularizer, and representation-learning objective in machine learning. Many works use persistent homology as a feature extractor for time series learning. For example, [35] compute persistence diagram statistics for classification, [50] uses Betti sequences as persistence-based features, and [28] combine time-delay embeddings with persistence vectorizations such as Betti curves and persistence landscapes. In contrast, we treat topology as complementary structure, augmenting learned latent representations rather than replacing them.

PersLay [15] is a neural layer that maps persistence diagrams to fixed-length vector representations from persistence diagrams for downstream machine learning models. Unlike PersLay, we use persistence vectorizations as supplementary features along the learned latent state, rather than the primary representation.

TopoCL incorporates topology into representation learning via contrastive learning objective [29]. On the other hand, we use explicit topological descriptors to augment learned latent representations for downstream prediction.

The JEPA family predicts latent representations rather than raw observations. I-JEPA [6] predicts latent targets for masked regions within images, while V-JEPA [5] extends this idea to masked regions in future video frames. These models pretrain general-purpose encoders on large external datasets, whereas we focus on lightweight dataset-specific encoders and study whether persistent topology features improve next latent forecasting on learned representations. We use V-JEPA only as an appendix robustness check (see Section C) to evaluate whether topology can also improve frozen pretrained representations.

2 Methodology

Our main method, shown in Fig. 2, embeds each input frame or state X_t at time step t into a latent vector z_t . For each t , we form a latent trajectory sliding window point cloud

$$P_t = \{z_{t-w+1}, \dots, z_t\},$$

where w is the sliding window length, and compute the VR persistence on P_t . The resulting topological descriptor B_t , such as H_0 or H_1 Betti curves or other persistence-vectorized features, is used as auxiliary information for next-latent state prediction. To combine z_t and B_t , we use early-fusion variants inspired by multimodal learning [8]. Before fusion, we standardize z_t and B_t separately using training-set statistics to prevent scale mismatch between the two feature groups. In direct concatenation, the predictor receives $[z_t, B_t]$. In learned fusion, a small Multi-Layer Perceptron (MLP)

maps $[z_t, B_t]$ to a fused representation $z_{\text{topo},t}$, which is then passed to the predictor.

Because Vietoris-Rips persistence is permutation-invariant, these descriptors depend on pairwise distances within P_t , but not on the temporal order of the latent states. So, the topological branch summarizes the geometry of the recent latent trajectory, while temporal order is modeled by the downstream LSTM predictor.

We emphasize H_1 descriptors in most controls because they capture loop-like structure, which is relevant for ring-shaped objects, holes, and recurring trajectories. Our stability diagnostic also shows that H_1 diagrams are less sensitive to input perturbations than H_0 diagrams in our latent point clouds, with $d_B(H_1)/d_H < d_B(H_0)/d_H$ across all dataset-encoder pairs (Appendix Table 6). We still include H_0 and $H_0 + H_1$ ablations to test this choice, rather than assume it.

We include a lightweight PersLay-style vectorization to obtain supplementary topological features for next latent prediction, rather than as the core prediction architecture.

We use a CNN encoder given its effectiveness in encoding sequential data and describing each frame’s contents [32], and a Long Short-Term Memory (LSTM) predictor for next-latent forecasting, as it is effective in learning time-based changes and handling sequential predictions [26]. For each dataset and seed, the encoder is trained then frozen; all topology and control modes are evaluated using the same frozen latent trajectories.

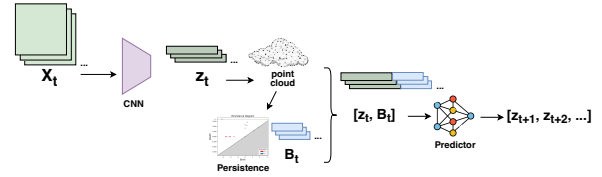


Figure 2: Diagram of the proposed method, which encodes each input X_t into a latent representation z_t , computes persistence features over a sliding latent window, concatenates them with z_t , and predicts future latent states

2.1 Why Topology Can Improve Prediction

Our topology descriptor B_t summarizes a sliding window of recent latent states and is provided as auxiliary input to the sequential predictor. The following proposition characterizes when this augmentation can reduce optimal prediction risk. Let $\mathcal{R}^*(U)$ denote the minimum expected mean-squared prediction error achievable using information U .

PROPOSITION 2.1. *Let H_t denote any representation of the latent history available to a predictor, and let B_t denote the corresponding persistent homology descriptor. Assume that the future target Y_t has finite second moment. Then the optimal prediction risks satisfy*

$$\mathcal{R}^*(H_t) - \mathcal{R}^*(H_t, B_t) = \mathbb{E} [\|\mathbb{E}[Y_t | H_t, B_t] - \mathbb{E}[Y_t | H_t]\|_2^2] \geq 0, \quad (1)$$

where Y_t is the future target.

PROOF. Let $m_H = \mathbb{E}[Y_t | H_t]$ and $m_{HB} = \mathbb{E}[Y_t | H_t, B_t]$. Since

$$Y_t - m_H = (Y_t - m_{HB}) + (m_{HB} - m_H),$$

taking squared norms and expectations yields

$$\mathbb{E}\|Y_t - m_H\|^2 = \mathbb{E}\|Y_t - m_{HB}\|^2 + \mathbb{E}\|m_{HB} - m_H\|^2 + 2\mathbb{E}[\langle Y_t - m_{HB}, m_{HB} - m_H \rangle].$$

The final term equals zero by the Pythagorean orthogonality property of conditional expectation [52]. Therefore,

$$\mathbb{E}\|Y_t - m_H\|^2 = \mathbb{E}\|Y_t - m_{HB}\|^2 + \mathbb{E}\|m_{HB} - m_H\|^2.$$

Rearranging proves the result. $\square$

Proposition 2.1 shows that adding topology cannot increase the optimal prediction error under squared loss. The improvement is positive when (B_t) changes the expected future target after accounting for the latent history represented by (H_t) , and zero otherwise. This population-level result concerns the optimal predictor and does not guarantee improvement for a finite model trained on finite data.

2.2 Autoencoder Variants

Our geometry-regularized autoencoder (GeoAE) is a convolutional autoencoder trained with reconstruction loss, plus a pairwise-distance preserving penalty. This penalty encourages normalized pairwise distances between latent codes to match normalized pairwise distances between the corresponding flattened input frames (ie, it penalizes distortions of the data’s metric structure within each mini-batch). This resembles the work of [17], though they use a continuous k-Nearest Neighbors graph objective, while we use a simpler mini-batch pairwise-distance penalty. The loss $\mathcal{L}_{\text{GeoAE}}$ is given by:

$$\mathcal{L}_{\text{GeoAE}} = \mathcal{L}_{\text{recon}}(x, \hat{x}) + \lambda \left\| \frac{D_X}{\overline{D_X}} - \frac{D_Z}{\overline{D_Z}} \right\|_F^2,$$

D_X is the pairwise Euclidean distance matrix between flattened input frames, D_Z is the pairwise Euclidean distance matrix between their latent vectors, and $\overline{D_X}$ and $\overline{D_Z}$ denote the mean entries of the respective corresponding distance matrices. The pairwise distance d between two vectors x_1 and x_2 is

$$d(x_1, x_2) = \sqrt{\sum_k (x_{1,k} - x_{2,k})^2}$$

The topology-regularized autoencoder (TopoAE) adds an H_0 persistence penalty between the input mini-batch X and its latent representation Z , following [37]:

$$\mathcal{L}_{\text{TopoAE}} = \mathcal{L}_{\text{recon}}(x, \hat{x}) + \lambda \mathcal{L}_{H_0}(X, Z)$$

Here, X denotes the mini-batch of flattened input frames, treated as a point cloud in pixel space, and Z denotes the corresponding latent codes. The term $\mathcal{L}_{H_0}(X, Z)$ compares the finite H_0 Vietoris-Rips persistence pairs of X and Z by matching the selected Euclidean pairwise distances at which connected components merge.

2.3 Baselines

For non-topological controls, we replace the persistence descriptor with Principal Component Analysis (PCA) [47] or Kernel PCA (KPCA) [42] summaries computed from the same sliding-window latent point cloud. PCA tests whether gains come from adding a linear summary of latent-window variance, while KPCA provides

a stronger nonlinear geometric control that captures curved latent structure without using persistence.

As non-topological manifold geometric controls, we compare against graph-spectral summaries computed from the same sliding-window latent point clouds used for persistence. For each window $P_t = \{z_{t-w+1}, \dots, z_t\}$, we build a weighted radial basis function (RBF) graph with affinities $W_{ij} = \exp(-\|z_i - z_j\|_2^2 / \sigma^2)$, where σ is chosen using the median pairwise-distance heuristic. We evaluate two manifold learning descriptors: 1) *Laplacian Eigenmaps*, using the smallest eigenvalues of the normalized graph Laplacian [11, 51], and 2) *Diffusion maps*, using the leading eigenvalues of the normalized diffusion operator [20], which summarize multiscale connectivity and diffusion geometry of the latent window.

As a dimension-matched nonlinear control, we compute random Fourier features (RFFs) from each standardized latent window [40], using the same fixed random map for training and testing. This tests whether generic nonlinear features explain the gains. Together, PCA/KPCA test linear and kernel variation, Laplacian and diffusion maps test manifold geometry, and RFFs test generic nonlinear expansion. These controls test whether improvements come from persistence-specific topology rather than generic linear, nonlinear, geometric, or manifold summaries of the latent window.

3 Experiments

3.1 Datasets

We run our methods on many sequential videos and time series datasets, both real and synthetic. The datasets, outlined in Table 1, are as follows:

- Bouncing disks/rings: synthetic videos with multiple filled disks, or hollow rings, undergoing irregular bouncing motion.
- Orbiting disks/rings: synthetic videos with filled disks, or hollow rings, following smoother periodic orbital motion.
- Lorenz-96: chaotic multivariate time series generated from the Lorenz-96 dynamical system [34] with 40 variables and forcing $F = 8$, rendered as 64×64 heatmap-like frame sequences.
- Moving MNIST [44]: videos of two moving handwritten digits.
- ElectricDevices [21]: real electric-device usage time series rendered as frame sequences.
- Glioblastoma and HeLa [36]: real microscopy videos from the Cell Tracking Challenge.

Some dataset frames are shown in Fig. 3.

3.2 Results

Experimental details are given in Section B. We predict the future frames’ latent representation and evaluate them using the Mean Square Error (MSE) and coefficient of determination R^2 . Results are averaged over 5 seeds with prediction horizon 5 and latent window size 15. Lower MSE and higher R^2 are better.

Table 2 reports the main latent forecasting results. Consistent with Proposition 2.1, direct topology augmentation improves over the z -only baseline in 24/27 dataset-encoder settings, suggesting that persistence descriptors provide complementary predictive

Table 1: Datasets used in the forecasting experiments. Sequence counts report the effective train/test clips used in the pipeline.

Dataset	Type	Modality	Sequence length	Frame size	Sequences	Source
Bouncing disks	Synthetic	Video	50	64×64	512/128	-
Bouncing rings	Synthetic	Video	50	64×64	512/128	-
Orbiting disks	Synthetic	Video	50	64×64	512/128	-
Orbiting rings	Synthetic	Video	50	64×64	512/128	-
Moving MNIST	Real	Video	20	64×64	1000/200	[link] [44]
HeLa	Real	Video	50	64×64	14/28	[link] [36]
Glioblastoma	Real	Video	50	64×64	128/64	[link] [36]
Lorenz-96	Synthetic	Time series	100	64×64	512/128	-
ElectricDevices	Real	Time series	96	32×32	1024/512	[link] [21]

Table 2: Main latent-forecasting results after selecting predictor hyperparameters on z-only validation MSE. Each dataset is evaluated with a standard AE, a geometry-regularized AE (GeoAE), and a topology-regularized AE (TopoAE). “Direct” denotes fixed topology descriptors appended to the latent state, while “Fusion” learns $z_{\text{topo}} = \text{MLP}([z, B])$ jointly with the predictor. Bold marks the best entry within each encoder row; * marks the best result for the dataset.

Dataset	Encoder	z only	Best direct z+topology	Best fusion
Bouncing disks	AE	0.3429±0.0266	0.3402±0.0256	0.3639±0.0297
	GeoAE	0.0013±0.0003	0.0010±0.0002*	0.0019±0.0007
	TopoAE	0.2158±0.0099	0.2070±0.0083	0.2259±0.0075
Bouncing rings	AE	0.2972±0.0474	0.2955±0.0495	0.3112±0.0561
	GeoAE	0.0013±0.0001	0.0011±0.0002*	0.0022±0.0002
	TopoAE	0.2104±0.0120	0.2024±0.0103	0.2160±0.0105
Orbiting disks	AE	0.0052±0.0018	0.0047±0.0016	0.0068±0.0027
	GeoAE	0.0017±0.0001	0.0011±0.0001*	0.0015±0.0001
	TopoAE	0.2281±0.0124	0.2083±0.0076	0.2761±0.0162
Orbiting rings	AE	0.0045±0.0014	0.0040±0.0013	0.0054±0.0020
	GeoAE	0.0017±0.0001	0.0010±0.0001*	0.0016±0.0002
	TopoAE	0.2321±0.0144	0.2254±0.0080	0.2617±0.0202
Moving MNIST	AE	0.0048±0.0019	0.0046±0.0018	0.0080±0.0030
	GeoAE	0.0009±0.0001*	0.0009±0.0001	0.0023±0.0005
	TopoAE	0.2868±0.0056	0.2748±0.0061	0.3037±0.0119
Lorenz-96	AE	0.1536±0.0057	0.1420±0.0119	0.1998±0.0223
	GeoAE	0.0012±0.0003*	0.0012±0.0001	0.0013±0.0003
	TopoAE	0.2726±0.0298	0.2745±0.0394	0.3264±0.0258
ElectricDevices	AE	0.0139±0.0072	0.0102±0.0031	0.0167±0.0094
	GeoAE	0.0014±0.0001	0.0011±0.0002*	0.0017±0.0002
	TopoAE	0.0615±0.0047	0.0492±0.0025	0.1120±0.0063
HeLa	AE	0.0437±0.0107	0.0383±0.0083	0.0496±0.0081
	GeoAE	0.0072±0.0032	0.0071±0.0033*	0.0133±0.0054
	TopoAE	0.0587±0.0365	0.0550±0.0324	0.0644±0.0351
Glioblastoma	AE	0.0027±0.0010	0.0021±0.0006	0.0071±0.0016
	GeoAE	0.0011±0.0001	0.0008±0.0001*	0.0063±0.0035
	TopoAE	0.0040±0.0013	0.0033±0.0008	0.0078±0.0015

structure beyond the compressed latent representation. GeoAE

gives the lowest MSE on all nine datasets, while direct augmentation consistently outperforms learned fusion. The complete results are reported in Table 13.

Table 3: Corrupted-feature capacity controls after z-only hyperparameter selection. Entries are mean MSE over five seeds; the full mean $\pm$ SD sweep is reported in Table ??. For each family, the best of zero, shuffle, noise, and temporal-shift variants is shown. **Bold marks the best entry within each encoder group. Lower is better.**

Dataset	AE				GeoAE				TopoAE			
	z	Lap.	Diff.	RFF	z	Lap.	Diff.	RFF	z	Lap.	Diff.	RFF
Bouncing disks	0.3434	0.3675	0.3642	0.3634	0.0013	0.0021	0.0020	0.0021	0.2160	0.2206	0.2222	0.2218
Bouncing rings	0.2957	0.3094	0.3120	0.3076	0.0013	0.0022	0.0019	0.0022	0.2103	0.2130	0.2146	0.2139
Orbiting disks	0.0052	0.0069	0.0069	0.0069	0.0017	0.0017	0.0016	0.0017	0.2280	0.2677	0.2630	0.2668
Orbiting rings	0.0045	0.0056	0.0055	0.0055	0.0017	0.0017	0.0016	0.0016	0.2321	0.2631	0.2702	0.2623
Moving MNIST	0.0048	0.0084	0.0086	0.0083	0.0009	0.0023	0.0023	0.0023	0.2868	0.3045	0.3027	0.3017
Lorenz-96	0.1534	0.1983	0.1984	0.1984	0.0011	0.0014	0.0014	0.0013	0.2717	0.3182	0.3196	0.3140
ElectricDevices	0.0138	0.0156	0.0153	0.0151	0.0015	0.0015	0.0019	0.0019	0.0612	0.1086	0.1067	0.1043
HeLa	0.0437	0.0486	0.0507	0.0461	0.0073	0.0115	0.0116	0.0109	0.0581	0.0634	0.0629	0.0638
Glioblastoma	0.0027	0.0066	0.0065	0.0077	0.0011	0.0059	0.0049	0.0051	0.0042	0.0084	0.0086	0.0080

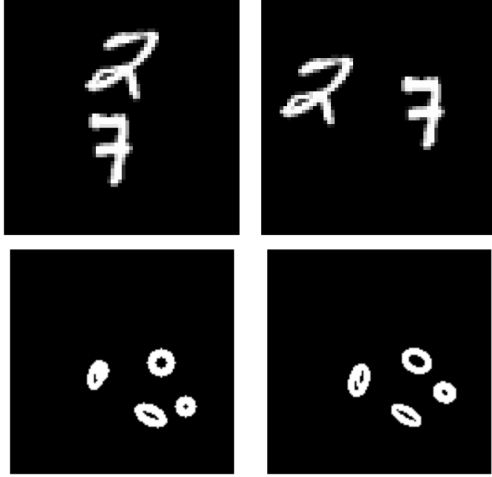


Figure 3: Example frames from the Moving MNIST (top) and bouncing rings (bottom) datasets.

Table 3 reports corrupted Laplacian, diffusion, and RFF capacity controls. Even after selecting the best corruption, these controls outperform z-only in only 2/81 settings, suggesting that additional dimensions or fusion capacity alone do not explain the gains

Table 4 compares topology with PCA, KPCA, Laplacian eigenmaps, diffusion maps, and dimension-matched RFF controls computed from the same latent windows. Topology outperforms these controls in most dataset–encoder settings, suggesting that the gains are not explained solely by linear variation, manifold geometry, or nonlinear expansion.

Figure 4 visualizes the improvements in MSE under various encoders and feature combinations, of Tables 2 and 4. Topology improves over the z-only baseline in 24/27 dataset–encoder settings and outperforms the Laplacian, diffusion, and RFF controls in 26/27, 23/27, and 26/27 settings, respectively. GeoAE gives the lowest topology-augmented MSE on all nine datasets.

Appendix Table 11 reports results using frozen V-JEPA2 features. Although frozen V-JEPA2 performs worse than the dataset-specific

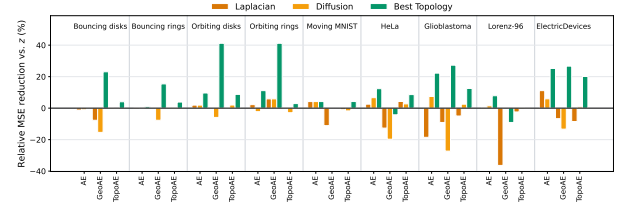


Figure 4: Relative MSE reduction over z-only for the best Laplacian, diffusion, RFF, and topology variants across datasets and encoders

encoders in this setting, topology consistently improves over z-only forecasting on all but one datasets.

These controls test whether performance gains come from meaningful topological information or simply from giving the predictor additional numerical features. Particularly, the random, shuffled, shifted, and zeroed settings help separate topology-specific information from generic feature augmentation.

Table 12 reports the best-performing topological descriptor for each dataset under each encoder variant.

4 Discussion

Latent representations do not necessarily preserve the visible topological structure of their original input; for example, a loop in pixel space may be compressed or distorted by the encoder, especially when the latter is trained only for reconstruction. This explains why H_1 is not always the strongest latent-space signal, as loop information may not survive the encoding. In contrast, H_0 captures connectivity and coarse clustering structure, which is often more stable in latent space. This motivates both the TopoAE regularizer, following [37], and the use of multiple topology descriptors, rather than assuming that pixel-space topology is preserved automatically.

GeoAE performs strongly because its distance-preserving regularizer encourages the latent space to retain input space geometry. This may provide the predictor with a more useful coordinate system and make topology computed over the latent trajectory more

Table 4: Latent-window controls after z-only hyperparameter selection. PCA, KPCA, graph-Laplacian, diffusion-map, and dimension-matched random Fourier features replace the persistence descriptor with non-topological summaries computed from the same latent window. “Best topology” is the best persistence descriptor among the evaluated direct and fused topology variants. Lower MSE is better.

Dataset	Encoder	z	Fuse PCA	Fuse KPCA	Best RFF	Best spectral	Fuse H_1	Best topology
Bouncing disks	AE	0.3429±0.0266	0.4270±0.0273	0.3644±0.0277	0.3437±0.0282	0.3453±0.0291 (Diffusion)	0.3639±0.0297	0.3402±0.0256 (H_1)
	GeoAE	0.0013±0.0003	0.0022±0.0007	0.0025±0.0003	0.0014±0.0003	0.0014±0.0002 (Laplacian)	0.0025±0.0003	0.0010±0.0002 (Landscape H_0)
	TopoAE	0.2158±0.0099	0.2258±0.0100	0.2217±0.0096	0.2177±0.0091	0.2152±0.0079 (Laplacian)	0.2259±0.0075	0.2070±0.0083 (Landscape $H_0 + H_1$)
Bouncing rings	AE	0.2972±0.0474	0.3902±0.0620	0.3104±0.0557	0.2937±0.0467	0.2986±0.0509 (Laplacian)	0.3112±0.0561	0.2955±0.0495 (Landscape H_1)
	GeoAE	0.0013±0.0001	0.0022±0.0003	0.0023±0.0003	0.0013±0.0002	0.0013±0.0003 (Laplacian)	0.0023±0.0002	0.0011±0.0002 (Landscape H_0)
	TopoAE	0.2104±0.0120	0.2187±0.0153	0.2148±0.0132	0.2082±0.0121	0.2096±0.0091 (Laplacian)	0.2160±0.0105	0.2024±0.0103 (PersLay H_0)
Orbiting disks	AE	0.0052±0.0018	0.0064±0.0023	0.0069±0.0027	0.0051±0.0015	0.0051±0.0016 (Laplacian)	0.0070±0.0028	0.0047±0.0016 (PersLay H_0)
	GeoAE	0.0017±0.0001	0.0016±0.0001	0.0016±0.0001	0.0017±0.0001	0.0017±0.0001 (Fuse Laplacian)	0.0015±0.0001	0.0011±0.0001 (PersLay H_0)
	TopoAE	0.2281±0.0124	0.2766±0.0071	0.2800±0.0105	0.2239±0.0079	0.2235±0.0107 (Diffusion)	0.2761±0.0162	0.2083±0.0076 (Landscape H_1)
Orbiting rings	AE	0.0045±0.0014	0.0055±0.0019	0.0056±0.0021	0.0047±0.0014	0.0044±0.0015 (Laplacian)	0.0056±0.0019	0.0040±0.0013 (PersLay H_0)
	GeoAE	0.0017±0.0001	0.0014±0.0001	0.0016±0.0001	0.0016±0.0001	0.0016±0.0001 (Fuse Laplacian)	0.0017±0.0001	0.0010±0.0001 (PersLay H_0)
	TopoAE	0.2321±0.0144	0.2723±0.0302	0.2653±0.0232	0.2337±0.0106	0.2326±0.0160 (Laplacian)	0.2617±0.0202	0.2254±0.0080 (Landscape H_1)
Moving MNIST	AE	0.0048±0.0019	0.0081±0.0033	0.0086±0.0028	0.0048±0.0020	0.0046±0.0019 (Laplacian)	0.0087±0.0028	0.0046±0.0018 ($H_0 + H_1$)
	GeoAE	0.0009±0.0001	0.0015±0.0005	0.0021±0.0005	0.0011±0.0002	0.0009±0.0001 (Diffusion)	0.0027±0.0005	0.0009±0.0001 (H_0)
	TopoAE	0.2868±0.0056	0.3000±0.0098	0.3041±0.0104	0.2903±0.0056	0.2895±0.0096 (Laplacian)	0.3037±0.0119	0.2748±0.0061 (Landscape $H_0 + H_1$)
Lorenz-96	AE	0.1536±0.0057	0.2081±0.0274	0.2042±0.0229	0.1441±0.0078	0.1505±0.0031 (Diffusion)	0.1998±0.0223	0.1420±0.0119 (PI H_1)
	GeoAE	0.0012±0.0003	0.0022±0.0003	0.0017±0.0003	0.0015±0.0001	0.0012±0.0002 (Diffusion)	0.0017±0.0005	0.0012±0.0001 (PersLay $H_0 + H_1$)
	TopoAE	0.2726±0.0298	0.4258±0.0256	0.3299±0.0254	0.2749±0.0300	0.2726±0.0309 (Diffusion)	0.3264±0.0258	0.2745±0.0394 (H_1)
ElectricDevices	AE	0.0139±0.0072	0.0172±0.0106	0.0170±0.0095	0.0124±0.0065	0.0118±0.0051 (Laplacian)	0.0167±0.0094	0.0102±0.0031 (PI H_1)
	GeoAE	0.0014±0.0001	0.0014±0.0002	0.0017±0.0002	0.0014±0.0001	0.0016±0.0003 (Laplacian)	0.0017±0.0001	0.0011±0.0002 ($H_0 + H_1$)
	TopoAE	0.0615±0.0047	0.1205±0.0105	0.1070±0.0167	0.0626±0.0080	0.0604±0.0053 (Diffusion)	0.1210±0.0144	0.0492±0.0025 (Landscape $H_0 + H_1$)
HeLa	AE	0.0437±0.0107	0.0480±0.0076	0.0488±0.0098	0.0418±0.0088	0.0409±0.0051 (Diffusion)	0.0496±0.0081	0.0383±0.0083 (Landscape H_1)
	GeoAE	0.0072±0.0032	0.0130±0.0044	0.0127±0.0041	0.0080±0.0033	0.0079±0.0038 (Laplacian)	0.0146±0.0033	0.0071±0.0033 (Landscape H_1)
	TopoAE	0.0587±0.0365	0.0640±0.0354	0.0675±0.0392	0.0605±0.0361	0.0558±0.0326 (Laplacian)	0.0644±0.0351	0.0550±0.0324 (Landscape H_1)
Glioblastoma	AE	0.0027±0.0010	0.0079±0.0025	0.0086±0.0022	0.0022±0.0005	0.0025±0.0009 (Diffusion)	0.0100±0.0024	0.0021±0.0006 (PersLay H_1)
	GeoAE	0.0011±0.0001	0.0066±0.0029	0.0062±0.0019	0.0017±0.0005	0.0012±0.0005 (Laplacian)	0.0063±0.0035	0.0008±0.0001 (Landscape H_0)
	TopoAE	0.0040±0.0013	0.0092±0.0031	0.0081±0.0017	0.0048±0.0015	0.0037±0.0011 (Diffusion)	0.0078±0.0015	0.0033±0.0008 (PersLay H_1)

informative. TopoAE is weaker in our setting because preserving coarse H_0 persistence does not necessarily preserve the metric and temporal details needed for next-latent prediction.

Across most datasets, GeoAE also exhibits the largest relative reduction from its z-only baseline after topology is added. This suggests that geometry regularization and persistent topology are complementary: organizing the latent points through pairwise-distance preservation may produce sliding-window point clouds whose topological summaries are more informative for forecasting.

We compare scenarios against the baseline case of inputting only z into the predictor. When using the Betti sequence as the only input to the predictor (without z), it consistently produces the weakest predictor of future latent states. This is expected because Betti sequences summarize connectivity and loop structure but omit important information, like position and intensity, needed to reconstruct the future latent vector.

One important limitation is that Vietoris-Rips persistence on a point cloud is permutation-invariant: it depends on pairwise distances between points, not on the order in which the latent states occur. This still allows temporal prediction to be possible, because a descriptor is computed for each sliding window and the LSTM models how these window-level summaries evolve over time. The TDA descriptor contributes an order-invariant summary of the recent latent trajectory, while the recurrent predictor retains the temporal ordering between successive windows. This may also explain why shuffled topology descriptors can remain competitive: shuffling removes their alignment with individual examples or time steps, but preserves aspects of their overall marginal distribution and scale.

Non-topological controls test whether the gains are explained by generic summaries of the same latent window dimension. PCA and KPCA test linear and kernel variation, Laplacian eigenmaps and diffusion maps test manifold geometry, and dimension-matched RFFs test generic nonlinear expansion. These controls sometimes improve prediction, showing that latent spread and geometry are informative. However, topology remains stronger across most dataset-encoder settings, especially when landscapes or PersLay-style vectorizations are used, suggesting that the gains are not explained solely by adding dimensions or summarizing latent variation.

There is no universal ranking among the persistence vectorizations. Betti curves are coarse counts over filtration values, while persistence images, landscapes, and PersLay-style features retain different aspects of the birth-death distribution. Betti curves perform well in several individual settings, while landscapes and PersLay give the most consistent advantages against the stronger matched controls. This suggests that retaining more birth-death structure can help, though the best vectorizations are dataset- and encoder-dependent.

As an additional robustness check, Appendix Table 11 evaluates frozen V-JEPA2 features. This is a much larger general-purpose video encoder than our dataset-specific autoencoders, and is not optimized for low-dimensional latent forecasting setting. Despite that, adding topology improves its forecasting performance, suggesting that the benefit of such auxiliary features is not specific to our encoder architecture.

Finally, the full sweep in Table 13 shows that topology-only inputs are consistently the weakest settings. Topological descriptors

summarize structure but do not contain enough appearance, position, or intensity information to forecast the next latent state on their own. The strongest results come from combining topology with z , supporting the view that topology provides complementary context, rather than a standalone representation.

We complement our experimental results with stability, recoverability, significance, and explainability analyses in Appendix A. The stability testing in Appendix A.2 shows that GeoAE gives the lowest H_0 ratio in 7/9 datasets and the lowest H_1 in 6/9 datasets. Its persistence diagrams therefore generally change by a smaller fraction of the latent point cloud displacement under input noise. This can occur because GeoAE regularizes pairwise distances, allowing latent coordinates to move while preserving enough relative geometry for VR persistence to remain comparatively stable. Because these quantities are normalized by d_H , they measure relative rather than absolute diagram stability.

The recoverability analysis in Appendix A.3 shows that for most datasets, H_1 descriptors are not reliably predictable from z . Explicit topology may therefore provide information that is difficult to access from the latent coordinates alone. Glioblastoma is the exception, while ElectricDevices and HeLa show partial recoverability. Thus, topology can help either by adding information that is hard to recover from z , or by making structure already present in z directly available to the predictor.

The significance analysis in Appendix A.4 compares fixed topology descriptors with fixed PCA, KPCA, Laplacian, diffusion, and RFF controls. All topology descriptors strongly outperform PCA and KPCA, while landscapes and PersLay provide consistent advantages over the Laplacian and diffusion controls. RFF is the strongest control: the advantages of landscapes and PersLay remain smaller but significant under the one-sided tests. These results suggest that generic nonlinear expansion explains part, but not all, of the observed gains. The corrupted-feature capacity controls in Appendix D further show that arbitrary auxiliary features rarely improve over z -only.

Lastly, the explainability analysis in Appendix A.5 compares two-component PCA explainability with forecasting MSE. The percentage above each PCA panel in Figure 6 shows a positive within-dataset association between two-component PCA explainability and forecasting MSE. GeoAE generally has lower PCA explainability but lower MSE, suggesting that its forecast-relevant information is distributed across more latent dimensions, rather than concentrated in a visually compact two-dimensional representation.

4.1 Limitations

One limitation of incorporating topology is the cost of computing persistence features over sliding windows. Appendix Table 5 shows that persistence is slower than other controls, though it adds little accelerator memory overhead. GeoAE also requires additional pairwise distance regularization during training, but uses the same encoder architecture and parameter count as the standard AE at inference time.

Another limitation is that the gains depend on the encoder, dataset, persistence vectorization method, and window parameters, which adds components that require validation. Learned fusion does not consistently outperform direct augmentation in our experiments, meaning it may not always be beneficial.

Our conclusions are limited to 3 encoder families, 9 datasets, and next-latent MSE. VR persistence is also permutation-invariant within each window and so summarizes trajectory geometry without preserving its internal temporal order. Future work should evaluate larger encoder families, longer forecasting horizons or different tasks, decoded-frame quality, and more scalable or temporally aware topology constructions.

5 Conclusion

In this paper, we show that persistent homology descriptors provide complementary predictive structure for next latent forecasting. Across 9 datasets and 3 encoder families, such augmentations improve over the z baseline in 24/27 dataset-encoder settings, while topology-only inputs consistently perform worst. Thus, topology is most useful as auxiliary context rather than as replacement for the latent space. Geometry-preserving encoding also gives the strongest overall forecasting performance, suggesting that latent geometry and persistent homology provide complementary benefits.

Matched PCA, KPCA, Laplacian, diffusion maps, and RFF controls show that the gains are not explained solely by additional dimensions manifold geometry, or generic nonlinear expansion. Stability and recoverability analyses suggest that the descriptors are relatively robust and are often not easily predicted from z alone. Significance tests show that the advantage depends on the vectorization: Betti curves remain competitive, while persistence landscapes and PersLay-style representations provide the most consistent improvements against the strongest controls.

Future work could explore many directions. One could consider more scalable or temporally aware persistence constructions, alternative filtrations such as sublevel-set persistence, and other fusion strategies. Different architectures and predictors can be used, such as transformers, across additional modalities and tasks such as classification and anomaly detection. Finally, encoder regularization, latent dimension, topology-aware losses, distance functions, and decoded-frame quality are directions that would be useful to assess the best use-case of topological context.

The article template’s documentation, available at <https://www.acm.org/publications/proceedings-template>, has a complete explanation of these commands and tips for their effective use.

Note that authors’ addresses are mandatory for journal articles.

Ethics and Privacy Statement

Our work poses little to no societal risk that might result from its publication. The societal impacts are minimal, beyond those considered by institutional review boards.

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A Additional Tests

A.1 Profiling

We profile inference-time cost using previously-trained encoders and predictors, discarding one warmup run and averaging over 3 measured repeats. Profile size 128 means each run is capped at 128 training sequences and 128 test sequences. Table 5 shows that topology overhead is dominated by the persistence phase: PCA/KPCA add approximately 0.3-0.5 seconds, RFF adds 0.2 seconds, and Laplacian and diffusion maps add 0.6-0.7 seconds, whereas persistence-based inputs add 1.7-2.1 seconds per run. Accelerator-memory overhead is small and runtime variance is mostly due to dataset size and sequence length.

This section contains additional tests to support the experiments, mainly the stability diagnosis, the recoverability, and the significance test.

A.2 Stability Diagnosis

Our topological descriptors are computed from sliding-window point clouds of latent vectors using VR persistence. Persistence diagrams are stable summaries of topology under suitable perturbations [19]; in our setting, where diagrams are computed from latent point clouds using VR filtrations, the corresponding point-cloud stability result relates Hausdorff perturbations of the latent cloud to bottleneck perturbations of the diagram (ie, if two point clouds are close in Hausdorff distance, then their persistence diagrams are close in bottleneck distance) [16]. For a latent point cloud P , let $\text{Dgm}_k(P)$ denote the k -dimensional persistence diagram obtained from its VR filtration. This does not imply stability of the full forecasting pipeline, but implies that the persistence step itself does not arbitrarily amplify small perturbations of the latent point cloud. For finite latent point clouds $P, Q \subset \mathbb{R}^d$, we use the Hausdorff distance d_H [12], which measures how far two point clouds are from each other

$$d_H(P, Q) = \max \left\{ \max_{p \in P} \min_{q \in Q} \|p - q\|_2, \max_{q \in Q} \min_{p \in P} \|q - p\|_2 \right\},$$

where $\|\cdot\|_2$ is Euclidean distance in latent space. The bottleneck distance d_B between two persistence diagrams $\text{Dgm}_k(P)$ and $\text{Dgm}_k(Q)$ is [23]

$$d_B(\text{Dgm}_k(P), \text{Dgm}_k(Q)) = \inf_{\gamma: \text{Dgm}_k(P) \rightarrow \text{Dgm}_k(Q)} \sup_{x \in \text{Dgm}_k(P)} \|x - \gamma(x)\|_\infty,$$

where γ ranges over matchings of birth-death points, allowing unmatched points to be matched to the diagonal. For Vietoris-Rips persistence, point-cloud stability gives

$$d_B(\text{Dgm}_k(P), \text{Dgm}_k(Q)) \leq C d_H(P, Q),$$

for homology dimension k , up to a filtration-dependent constant C [16].

To test this stability empirically, we perturb the input frames X with additive noise and compare the clean and noisy latent point clouds. Let $P_t = \{z_{t-w+1}, \dots, z_t\}$ be the latent window from clean frames and P'_t the corresponding window after noising the input frames and re-encoding them. We measure the latent perturbation by the Hausdorff distance $d_H(P_t, P'_t)$, and the topological perturbation by the bottleneck distance $d_B(\text{Dgm}_k(P_t), \text{Dgm}_k(P'_t))$. We

report the ratio

$$\rho_k(t) = \frac{d_B(\text{Dgm}_k(P_t), \text{Dgm}_k(P'_t))}{d_H(P_t, P'_t)}.$$

Values $\rho_k(t) < 1$ indicate that the persistence diagram changes less than the latent point cloud does, under the same perturbation.

Appendix Table 6 evaluates the stability of the latent persistence branch under input noise σ . When $\sigma = 0.10$, all reported ratios are below 1, meaning persistence diagrams change less than the latent point clouds themselves. In particular, the H_1 ratios $\rho_1 = d_B(\text{Dgm}_1(P_t), \text{Dgm}_1(P'_t))/d_H(P_t, P'_t)$ are consistently smaller than the H_0 ratios, suggesting that loop-level structure is less sensitive to input noise than connected-component structure in these latent point clouds.

GeoAE often exhibits larger latent point cloud displacement d_H under input noise, but yields the lowest relative H_0 and H_1 changes in 7/9 and 6/9 datasets respectively. So, its persistence diagrams generally change by a smaller fraction of the corresponding latent displacement. This may occur because GeoAE regularizes pairwise distances in latent space: noise can move the latent coordinates, increasing (d_H) , while preserving much of the relative geometry of the point cloud. Since VR persistence depends on pairwise distances rather than absolute coordinates, the topology can remain relatively stable even when the latent vectors move. These ratios measure stability relative to (d_H) , rather than absolute diagram change.

A.3 Recoverability Testing

We run a diagnostic probe experiment [3] to test whether the topological descriptor H_1 is recoverable from the latent vector z . For each encoder family, we freeze the learned encoder, take latent vectors z , and train a probe, either ridge regression (linear) or MLP (nonlinear) with 1 hidden layer, to predict the corresponding H_1 topology vector computed from the sliding-window latent point cloud. The probe is trained on the training split and evaluated on held-out test sequences, with R^2 measuring how well z alone predicts the topology feature.

We freeze the encoder and train probes from z to the H_1 topology vector. The linear probe is ridge regression with $\alpha \in \{0.1, 1, 10, 100, 10^3, 10^4, 10^5\}$, selected on a validation split. The nonlinear probe is a one-hidden-layer MLP with hidden size 128, ReLU activation, learning rate 10^{-3} , weight decay 10^{-4} , and a maximum of 300 iterations. For both probes, we use a sequence-level random held-out split with 20% test sequences and 20% validation inside the training set, standardize z and the target topology vector, remove near-constant topology bins with train-set standard deviation $\leq 10^{-3}$, and report held-out R^2 over 5 seeds.

The probe results, which are presented in Table 7, show that for most datasets, the H_1 descriptors are not reliably predictable from z , implying that injecting explicit topology provides information that is not easily accessible from the latent coordinates alone. The exception is Glioblastoma, where R^2 is high for all encoders; ElectricDevices and HeLa also show partial recoverability. This does not contradict our thesis: topology can help either by adding information that is difficult to recover from z , or by making structure that is already partly present in z directly available to the predictor. On several datasets, the nonlinear MLP probe underperforms the linear ridge probe, especially for GeoAE on the motion datasets.

Table 5: GPU profiling summary at profile size 128, averaged over datasets after discarding profiling warmup runs. Runtime, added time, peak CPU RSS, peak accelerator memory, and phase times are reported as mean $\pm$ std across datasets and measured repeats. Added time and Runtime/ z are relative to the corresponding z -only run within the same dataset, encoder family, and profiling repeat. Persistence is CPU-side in our implementation.

Encoder	Input mode	Runtime (s)	Added time (s)	Runtime/ z	Peak memory (MB)		Phase time (s)			
					CPU RSS	Accel.	Enc.	Feature/fuse	Persist.	Pred.
AE	z	0.05 $\pm$ 0.02	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	4047 $\pm$ 2020	134 $\pm$ 19	0.03 $\pm$ 0.02	–	–	0.00 $\pm$ 0.00
AE	z + PCA	0.36 $\pm$ 0.24	0.31 $\pm$ 0.22	7.00 $\pm$ 4.28	4051 $\pm$ 2014	134 $\pm$ 19	0.03 $\pm$ 0.02	0.30 $\pm$ 0.22	–	0.00 $\pm$ 0.00
AE	z + KPCA	0.51 $\pm$ 0.35	0.46 $\pm$ 0.33	10.05 $\pm$ 6.31	4051 $\pm$ 2014	134 $\pm$ 19	0.03 $\pm$ 0.02	0.46 $\pm$ 0.33	–	0.00 $\pm$ 0.00
AE	z_{fuse} + H_1	1.74 $\pm$ 1.19	1.69 $\pm$ 1.17	33.81 $\pm$ 23.22	4057 $\pm$ 2008	138 $\pm$ 18	0.03 $\pm$ 0.02	0.00 $\pm$ 0.00	1.31 $\pm$ 0.90	0.01 $\pm$ 0.00
AE	z_{fuse} + PI	2.02 $\pm$ 1.37	1.97 $\pm$ 1.35	39.40 $\pm$ 26.88	4058 $\pm$ 2007	139 $\pm$ 19	0.03 $\pm$ 0.02	0.28 $\pm$ 0.18	1.31 $\pm$ 0.90	0.01 $\pm$ 0.00
AE	z_{fuse} + Landscape	1.86 $\pm$ 1.27	1.82 $\pm$ 1.25	36.29 $\pm$ 24.56	4058 $\pm$ 2007	140 $\pm$ 19	0.03 $\pm$ 0.02	0.13 $\pm$ 0.08	1.30 $\pm$ 0.90	0.01 $\pm$ 0.00
AE	z_{fuse} + PersLay	1.82 $\pm$ 1.21	1.77 $\pm$ 1.20	35.48 $\pm$ 23.80	4058 $\pm$ 2007	140 $\pm$ 19	0.03 $\pm$ 0.02	0.08 $\pm$ 0.06	1.31 $\pm$ 0.89	0.01 $\pm$ 0.00
AE	z_{fuse} + PCA	0.35 $\pm$ 0.24	0.31 $\pm$ 0.22	6.98 $\pm$ 4.30	4058 $\pm$ 2007	138 $\pm$ 18	0.03 $\pm$ 0.02	0.30 $\pm$ 0.22	–	0.01 $\pm$ 0.00
AE	z_{fuse} + KPCA	0.51 $\pm$ 0.35	0.47 $\pm$ 0.33	10.05 $\pm$ 6.31	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.46 $\pm$ 0.33	–	0.01 $\pm$ 0.00
AE	z_{fuse} + Laplacian	0.72 $\pm$ 0.49	0.67 $\pm$ 0.47	14.03 $\pm$ 8.64	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.66 $\pm$ 0.47	–	0.01 $\pm$ 0.00
AE	z_{fuse} + Diffusion	0.69 $\pm$ 0.47	0.64 $\pm$ 0.45	13.33 $\pm$ 8.38	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.63 $\pm$ 0.45	–	0.01 $\pm$ 0.00
AE	z_{fuse} + RFF	0.25 $\pm$ 0.16	0.20 $\pm$ 0.14	4.93 $\pm$ 2.72	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.20 $\pm$ 0.14	–	0.01 $\pm$ 0.00
GeoAE	z	0.05 $\pm$ 0.02	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	4061 $\pm$ 2003	134 $\pm$ 19	0.03 $\pm$ 0.02	–	–	0.00 $\pm$ 0.00
GeoAE	z + PCA	0.36 $\pm$ 0.24	0.31 $\pm$ 0.22	6.95 $\pm$ 4.40	4061 $\pm$ 2003	134 $\pm$ 19	0.03 $\pm$ 0.02	0.31 $\pm$ 0.22	–	0.00 $\pm$ 0.00
GeoAE	z + KPCA	0.52 $\pm$ 0.36	0.47 $\pm$ 0.34	9.95 $\pm$ 6.53	4061 $\pm$ 2003	134 $\pm$ 19	0.03 $\pm$ 0.02	0.47 $\pm$ 0.34	–	0.00 $\pm$ 0.00
GeoAE	z_{fuse} + H_1	1.81 $\pm$ 1.24	1.76 $\pm$ 1.22	34.80 $\pm$ 25.45	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.00 $\pm$ 0.00	1.37 $\pm$ 0.95	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + PI	2.11 $\pm$ 1.42	2.06 $\pm$ 1.40	40.80 $\pm$ 29.51	4061 $\pm$ 2003	139 $\pm$ 19	0.03 $\pm$ 0.02	0.30 $\pm$ 0.19	1.37 $\pm$ 0.94	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + Landscape	1.95 $\pm$ 1.32	1.91 $\pm$ 1.30	37.77 $\pm$ 27.16	4061 $\pm$ 2003	140 $\pm$ 19	0.03 $\pm$ 0.02	0.14 $\pm$ 0.09	1.37 $\pm$ 0.94	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + PersLay	1.96 $\pm$ 1.29	1.91 $\pm$ 1.27	38.01 $\pm$ 26.94	4061 $\pm$ 2003	139 $\pm$ 19	0.03 $\pm$ 0.02	0.14 $\pm$ 0.10	1.40 $\pm$ 0.95	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + PCA	0.36 $\pm$ 0.24	0.31 $\pm$ 0.22	7.01 $\pm$ 4.41	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.31 $\pm$ 0.22	–	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + KPCA	0.52 $\pm$ 0.35	0.47 $\pm$ 0.33	9.97 $\pm$ 6.47	4061 $\pm$ 2003	138 $\pm$ 18	0.03 $\pm$ 0.02	0.47 $\pm$ 0.33	–	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + Laplacian	0.72 $\pm$ 0.48	0.67 $\pm$ 0.46	13.70 $\pm$ 8.79	4064 $\pm$ 2000	138 $\pm$ 18	0.03 $\pm$ 0.02	0.66 $\pm$ 0.46	–	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + Diffusion	0.68 $\pm$ 0.46	0.63 $\pm$ 0.44	13.06 $\pm$ 8.15	4064 $\pm$ 2000	138 $\pm$ 18	0.03 $\pm$ 0.02	0.63 $\pm$ 0.44	–	0.01 $\pm$ 0.00
GeoAE	z_{fuse} + RFF	0.25 $\pm$ 0.15	0.20 $\pm$ 0.14	4.85 $\pm$ 2.68	4064 $\pm$ 2000	138 $\pm$ 18	0.03 $\pm$ 0.02	0.19 $\pm$ 0.14	–	0.01 $\pm$ 0.00
TopoAE	z	0.05 $\pm$ 0.02	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	4064 $\pm$ 2000	134 $\pm$ 19	0.03 $\pm$ 0.02	–	–	0.00 $\pm$ 0.00
TopoAE	z + PCA	0.35 $\pm$ 0.23	0.30 $\pm$ 0.22	6.81 $\pm$ 4.29	4064 $\pm$ 2000	134 $\pm$ 19	0.03 $\pm$ 0.02	0.30 $\pm$ 0.22	–	0.00 $\pm$ 0.00
TopoAE	z + KPCA	0.51 $\pm$ 0.35	0.46 $\pm$ 0.33	9.88 $\pm$ 6.67	4064 $\pm$ 2000	134 $\pm$ 19	0.03 $\pm$ 0.02	0.46 $\pm$ 0.33	–	0.00 $\pm$ 0.00
TopoAE	z_{fuse} + H_1	1.75 $\pm$ 1.21	1.70 $\pm$ 1.20	33.50 $\pm$ 24.59	4064 $\pm$ 2000	139 $\pm$ 19	0.03 $\pm$ 0.02	0.00 $\pm$ 0.00	1.31 $\pm$ 0.92	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + PI	2.00 $\pm$ 1.38	1.96 $\pm$ 1.36	38.51 $\pm$ 27.87	4064 $\pm$ 1999	139 $\pm$ 19	0.03 $\pm$ 0.02	0.28 $\pm$ 0.18	1.30 $\pm$ 0.91	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + Landscape	1.87 $\pm$ 1.29	1.82 $\pm$ 1.27	35.84 $\pm$ 26.15	4064 $\pm$ 1999	141 $\pm$ 20	0.03 $\pm$ 0.02	0.13 $\pm$ 0.08	1.30 $\pm$ 0.90	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + PersLay	1.82 $\pm$ 1.24	1.77 $\pm$ 1.22	35.05 $\pm$ 25.28	4065 $\pm$ 1999	139 $\pm$ 19	0.03 $\pm$ 0.02	0.08 $\pm$ 0.06	1.31 $\pm$ 0.91	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + PCA	0.36 $\pm$ 0.24	0.31 $\pm$ 0.22	6.86 $\pm$ 4.27	4068 $\pm$ 1995	138 $\pm$ 18	0.03 $\pm$ 0.02	0.30 $\pm$ 0.22	–	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + KPCA	0.52 $\pm$ 0.35	0.47 $\pm$ 0.33	9.93 $\pm$ 6.66	4071 $\pm$ 1992	138 $\pm$ 18	0.03 $\pm$ 0.02	0.46 $\pm$ 0.33	–	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + Laplacian	0.72 $\pm$ 0.49	0.67 $\pm$ 0.47	13.69 $\pm$ 9.01	4071 $\pm$ 1992	138 $\pm$ 18	0.03 $\pm$ 0.02	0.66 $\pm$ 0.47	–	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + Diffusion	0.69 $\pm$ 0.46	0.64 $\pm$ 0.45	13.13 $\pm$ 8.51	4072 $\pm$ 1991	138 $\pm$ 18	0.03 $\pm$ 0.02	0.64 $\pm$ 0.45	–	0.01 $\pm$ 0.00
TopoAE	z_{fuse} + RFF	0.25 $\pm$ 0.15	0.20 $\pm$ 0.13	4.79 $\pm$ 2.67	4079 $\pm$ 1983	138 $\pm$ 18	0.03 $\pm$ 0.02	0.19 $\pm$ 0.13	–	0.01 $\pm$ 0.00

This likely reflects overfitting or unstable optimization given limited training samples relative to MLP capacity, rather than proving that nonlinear recoverability is impossible. We therefore treat the MLP probe as redundancy diagnosis.

A.4 Significance Testing

To test whether topology improves over non-topological geometric controls, we compare fixed topology descriptors against fixed PCA, KPCA, Laplacian, diffusion-map, and RFF descriptors using paired MSE values. For each descriptor, we first average MSE over encoder families within each dataset, producing one paired comparison per dataset, then apply a one-sided Wilcoxon signed-rank test over the 9 datasets [53]. This avoids treating encoder variants as independent samples and avoids selecting the best method based on test performance.

Table 8, in conjunction with Fig. 4, shows that the topology advantage depends on both the persistence vectorization and the

control. All topology descriptors strongly outperform PCA and KPCA, while persistence landscapes and PersLay also outperform the Laplacian and diffusion controls. RFF is the strongest control: Betti curves and persistence images are not significantly better than RFF, while landscapes and PersLay retain smaller one-sided advantages ($p = 0.0273$) and ($p = 0.0371$). Thus, generic nonlinear expansion explains part of the gain, but richer persistence vectorizations provide the most consistent advantage across the matched controls.

A.5 Explainability

Each encoder transforms the same observations X into a distinct latent geometry. PCA explainability is the percentage of total latent variance captured by PC1 and PC2; a higher value indicates the latent representation is concentrated along fewer dominant directions. Note that this does not mean the dataset itself is simpler. Figure 6 shows the PCA components for each dataset and encoder.

Table 6: Latent persistence stability diagnostic at input noise level $\sigma = 0.10$. We add Gaussian noise to the input, re-encode the clean and perturbed sequences, and compare latent-window point clouds and Vietoris–Rips persistence diagrams. d_H is the Hausdorff distance between clean and perturbed latent point clouds, and $d_B(H_k)/d_H$ reports the relative diagram change. Values below one indicate that the persistence diagram changes less than the latent point cloud under the same perturbation.

Dataset	Encoder	$d_H(z, z')$	$d_B(H_0)$	$d_B(H_0)/d_H$	$d_B(H_1)$	$d_B(H_1)/d_H$
Bouncing rings	AE	0.813±0.145	0.470±0.065	0.582±0.037	0.103±0.017	0.128±0.014
	GeoAE	8.946±1.030	1.365±0.121	0.153±0.006	0.317±0.013	0.036±0.006
	TopoAE	0.954±0.057	0.532±0.030	0.558±0.018	0.115±0.007	0.121±0.003
Bouncing disks	AE	0.832±0.127	0.387±0.039	0.468±0.030	0.081±0.005	0.098±0.011
	GeoAE	10.151±1.404	1.660±0.229	0.164±0.011	0.353±0.017	0.035±0.007
	TopoAE	0.797±0.063	0.440±0.043	0.551±0.025	0.095±0.009	0.118±0.005
Orbiting rings	AE	2.562±0.454	1.274±0.119	0.503±0.044	0.376±0.010	0.150±0.028
	GeoAE	10.393±1.153	1.837±0.475	0.175±0.029	0.333±0.014	0.032±0.005
	TopoAE	1.046±0.066	0.535±0.036	0.512±0.014	0.224±0.009	0.215±0.008
Orbiting disks	AE	2.954±0.597	1.344±0.171	0.461±0.040	0.378±0.026	0.131±0.019
	GeoAE	11.621±1.551	2.283±0.717	0.192±0.040	0.372±0.015	0.033±0.006
	TopoAE	0.773±0.062	0.387±0.024	0.501±0.018	0.151±0.007	0.196±0.009
Moving MNIST	AE	2.898±0.385	1.248±0.178	0.432±0.040	0.166±0.027	0.058±0.008
	GeoAE	7.129±0.785	1.156±0.088	0.163±0.015	0.259±0.015	0.037±0.003
	TopoAE	0.780±0.064	0.370±0.028	0.475±0.010	0.070±0.013	0.089±0.010
Lorenz-96	AE	1.599±0.351	0.772±0.231	0.477±0.039	0.008±0.002	0.005±0.000
	GeoAE	12.208±0.784	5.477±0.069	0.450±0.034	0.269±0.020	0.022±0.003
	TopoAE	1.232±0.040	0.522±0.017	0.424±0.011	0.009±0.003	0.007±0.002
ElectricDevices	AE	8.154±0.512	3.315±0.055	0.408±0.025	0.256±0.028	0.031±0.002
	GeoAE	9.817±1.135	3.719±0.122	0.383±0.045	0.331±0.014	0.034±0.003
	TopoAE	7.141±0.413	3.299±0.113	0.463±0.026	0.302±0.026	0.042±0.003
Glioblastoma	AE	16.038±0.174	5.737±1.389	0.357±0.083	0.475±0.093	0.030±0.006
	GeoAE	16.045±0.109	5.507±0.196	0.343±0.013	0.431±0.044	0.027±0.003
	TopoAE	16.181±0.387	5.553±0.393	0.343±0.022	0.654±0.049	0.040±0.003
HeLa	AE	7.967±0.980	4.317±0.322	0.544±0.027	0.099±0.025	0.012±0.002
	GeoAE	15.730±0.149	3.610±0.431	0.230±0.029	0.387±0.018	0.025±0.001
	TopoAE	12.806±3.612	4.206±0.486	0.352±0.106	0.392±0.080	0.032±0.007

GeoAE generally has a lower PCA explainability but lower forecasting MSE, whereas TopoAE is often more concentrated in two dimensions yet predicts worse.

As summarized in Table 9, PCA explainability is positively associated with forecasting MSE in every dataset (Spearman $\rho = 0.5$ or 1.0) [43]. So, concentration into a few high-variance directions does not imply better predictability; GeoAE may instead distribute predictive information across more latent dimensions, though this association does not imply causality.

B Implementation Details

B.1 Hardware

We ran the experiments on a workstation with an Intel Core i7-13700 CPU, 30 GB RAM, and an NVIDIA GeForce RTX 4090 (24 GB VRAM) running Ubuntu 24.10 with CUDA 13.0 and PyTorch 2.13.0.

We also used an Apple MacBook Pro with an M4 Pro (14-core CPU) chip, 48 GB unified memory, running macOS 26.5.2 and PyTorch 2.8.0.

We compute Vietoris–Rips (VR) persistence of latent point clouds using Ripser [10], persistence image and persistence landscape vectorizations using GUDHI [46], and bottleneck distances in the stability diagnostic using Persim [41].

B.2 Model Architecture

All autoencoder variants use the same CNN encoder architecture, differing only in training objective. The encoder maps each grayscale frame $X_t \in \mathbb{R}^{1 \times H \times W}$ to a latent vector z_t , using two convolutional layers, $1 \rightarrow 16$ and $16 \rightarrow 32$, both with kernel size 5, stride 2, padding 2, and ReLU activation. The resulting feature map is reduced using adaptive average pooling to 4×4 , flattened, and projected with a linear layer to the dataset-specific latent dimension.

Table 7: Per-dataset $z \rightarrow H_1$ topology-probe R^2 on held-out examples. Values are mean $\pm$ standard deviation over seeds for both the linear ridge and nonlinear MLP probes.

Dataset	Linear ridge probe			Nonlinear MLP probe		
	AE	GeoAE	TopoAE	AE	GeoAE	TopoAE
Bouncing disks	0.035 $\pm$ 0.011	0.006 $\pm$ 0.006	0.026 $\pm$ 0.007	0.044 $\pm$ 0.018	-0.027 $\pm$ 0.022	0.038 $\pm$ 0.011
Bouncing rings	0.024 $\pm$ 0.013	-0.000 $\pm$ 0.001	0.036 $\pm$ 0.013	0.034 $\pm$ 0.018	-0.031 $\pm$ 0.011	0.045 $\pm$ 0.017
Orbiting disks	0.006 $\pm$ 0.004	0.004 $\pm$ 0.006	0.020 $\pm$ 0.013	0.032 $\pm$ 0.012	-0.022 $\pm$ 0.016	0.065 $\pm$ 0.013
Orbiting rings	-0.003 $\pm$ 0.005	-0.002 $\pm$ 0.004	0.013 $\pm$ 0.011	-0.008 $\pm$ 0.008	-0.031 $\pm$ 0.012	0.033 $\pm$ 0.014
Moving MNIST	-0.001 $\pm$ 0.010	-0.006 $\pm$ 0.007	0.014 $\pm$ 0.012	-0.006 $\pm$ 0.028	-0.085 $\pm$ 0.056	0.016 $\pm$ 0.027
Lorenz-96	-0.003 $\pm$ 0.002	-0.003 $\pm$ 0.005	0.000 $\pm$ 0.002	-0.024 $\pm$ 0.016	-0.171 $\pm$ 0.187	-0.013 $\pm$ 0.008
ElectricDevices	0.070 $\pm$ 0.054	0.229 $\pm$ 0.061	0.117 $\pm$ 0.046	0.125 $\pm$ 0.052	0.311 $\pm$ 0.062	0.224 $\pm$ 0.076
Glioblastoma	0.775 $\pm$ 0.050	0.764 $\pm$ 0.053	0.770 $\pm$ 0.021	0.958 $\pm$ 0.030	0.944 $\pm$ 0.113	0.870 $\pm$ 0.146
HeLa	0.048 $\pm$ 0.700	0.427 $\pm$ 0.143	0.427 $\pm$ 0.119	0.040 $\pm$ 0.363	0.581 $\pm$ 0.348	0.404 $\pm$ 0.381

Table 8: Fixed paired comparisons of persistence descriptors with non-topological controls. Wins are reported over all 27 dataset-encoder settings, whereas statistical tests use nine independent dataset averages to avoid treating encoder variants as independent samples. Lower MSE is better.

Topology descriptor	Wins	Avg. gain	Wilcoxon p	Two-sided p
Betti H_1 vs. PCA	25/27	25.9%	0.0020	0.0039
Betti H_1 vs. KPCA	26/27	21.5%	0.0020	0.0039
Betti H_1 vs. Laplacian	14/27	2.1%	0.0039	0.0078
Betti H_1 vs. Diffusion	14/27	0.6%	0.0195	0.0391
Betti H_1 vs. RFF	14/27	1.7%	0.1016	0.2031
PI H_1 vs. PCA	23/27	24.3%	0.0020	0.0039
PI H_1 vs. KPCA	24/27	20.0%	0.0020	0.0039
PI H_1 vs. Laplacian	13/27	-1.3%	0.1250	0.2500
PI H_1 vs. Diffusion	12/27	-3.1%	0.1250	0.2500
PI H_1 vs. RFF	12/27	-1.5%	0.4102	0.8203
Landscape H_1 vs. PCA	25/27	28.7%	0.0020	0.0039
Landscape H_1 vs. KPCA	26/27	24.5%	0.0020	0.0039
Landscape H_1 vs. Laplacian	18/27	6.4%	0.0020	0.0039
Landscape H_1 vs. Diffusion	16/27	5.1%	0.0059	0.0117
Landscape H_1 vs. RFF	14/27	6.1%	0.0273	0.0547
PersLay H_1 vs. PCA	24/27	27.5%	0.0020	0.0039
PersLay H_1 vs. KPCA	25/27	23.2%	0.0020	0.0039
PersLay H_1 vs. Laplacian	16/27	5.2%	0.0020	0.0039
PersLay H_1 vs. Diffusion	18/27	3.8%	0.0020	0.0039
PersLay H_1 vs. RFF	15/27	4.8%	0.0371	0.0742

We use latent dimension 48 for Moving MNIST, 64 for the synthetic video datasets, Lorenz-96, and ElectricDevices, and 128 for HeLa and Glioblastoma.

Unless otherwise stated, reconstruction uses an MLP decoder with Linear(d_z , 512), ReLU, Linear(512, HW), and sigmoid output. For forecasting, we use a one-layer LSTM followed by a linear projection to the latent dimension. The predictor hidden dimension is selected by validation on the z -only baseline and then fixed for all modes: 128 for Lorenz-96 and 256 for the remaining datasets. In learned-fusion modes, $[z_t, B_t]$ is passed through a fusion MLP with Linear-GELU-LayerNorm-Linear-GELU, producing a fused vector before the LSTM predictor.

For each dataset, we performed a small validation-based hyperparameter search using only the z -only latent forecasting baseline. We

varied the predictor learning rate over $[0.1, 0.3, 1, 3, 5, 10] \times 10^{-3}$, hidden dimension over $[64, 128, 256]$, and training epochs over $[8, 20, 40]$, selecting the configuration with the lowest validation MSE. The selected dataset-level configuration was then fixed and reused for all topology, PCA, KPCA, and fusion variants, so that improvements are not due to mode-specific hyperparameter tuning.

C Other Scenarios

This part includes additional ideas we explored.

C.1 3D scenario

We tested a spatiotemporal TDA variant, shown in Figure 5, where a sequence X is treated as a 3D volume with axes (t, y, x) . Instead of computing the persistence on a latent point cloud, we stack

Table 9: Two-component PCA explainability and z-forecasting MSE across encoders. Each PCA value is the percentage of latent variance explained by PC1 and PC2. The final column reports Spearman’s ρ

Dataset	AE		GeoAE		TopoAE		Spearman ρ
	PC1+PC2 (%)	MSE	PC1+PC2 (%)	MSE	PC1+PC2 (%)	MSE	
Bouncing disks	52.0	0.3429	21.3	0.0013	47.5	0.2158	1.00
Bouncing rings	75.8	0.2972	18.3	0.0013	49.9	0.2104	1.00
Orbiting disks	93.6	0.0052	13.7	0.0017	42.7	0.2281	0.50
Orbiting rings	94.3	0.0045	15.4	0.0017	44.6	0.2321	0.50
Moving MNIST	89.8	0.0048	35.8	0.0009	48.3	0.2868	0.50
Lorenz-96	33.2	0.1536	14.6	0.0012	25.3	0.2726	0.50
ElectricDevices	99.6	0.0139	29.4	0.0014	64.7	0.0615	0.50
Glioblastoma	49.8	0.0027	28.1	0.0011	59.9	0.0040	1.00
HeLa	97.9	0.0437	72.9	0.0072	86.8	0.0587	0.50



Figure 5: Visualization of the 3D construction of the Bouncing Rings dataset. Consecutive frames are stacked along the time axis to form a cubical volume, on which we compute H_0 , H_1 , and H_2 persistence features

consecutive input frames along the time axis and compute cubical persistence [27] directly on the resulting volume. For time series datasets, sequence values are arranged into 2D frame-like arrays, which are then stacked over time to form the 3D cubical volume. We then extract Betti curve features from H_0 , H_1 , and H_2 , and their combinations, and concatenate the selected features with the latents z for prediction. In this setting, H_0 tracks connected spatiotemporal components, H_1 tracks loop structures, and H_2 tracks enclosed voids or cavities in the volume.

We run this algorithm once per dataset, with results shown in Table 10. The best 3D TDA mode improves weighted MSE over the no-TDA baseline on all datasets, suggesting that volume-level topology can provide useful auxiliary signal. However, because this method computes cubical persistence in input space X , it is substantially slower, taking several hours to produce Table 10. We treat this as an exploratory extension, rather than a main method, since the latent point cloud approach gives similar gains at much lower computational cost. Its consistent improvement across datasets suggests that spatiotemporal H_2 and other persistence features may be useful in settings where the extra computational cost is acceptable.

C.2 V-JEPA

We also evaluate whether the topological signal helps when the latent representation comes from a large pretrained video encoder. We use the V-JEPA2 model ‘facebook/vjepa2-vitl-fpc64-256’ [link], and train only the downstream predictor and fusion head

on top of its latent features. Because V-JEPA is computationally expensive, we run this robustness check for a single run with `latent_tda_max_train = 64` and `latent_tda_max_test = 32`.

Results, presented in Table 11, show that adding topology improves frozen V-JEPA2 forecasting on 8/9 datasets, with an average MSE reduction of roughly 4.3% over raw V-JEPA2 latents. We treat this as a robustness check rather than a main benchmark: V-JEPA2 is substantially heavier than our dataset-specific encoders and is unnecessary for our low-dimensional forecasting setting, but the result suggests that topological auxiliary features can improve latent forecasting even when the representation comes from a large frozen pretrained video model.

D Additional Runs

Appendix Table 3 evaluates corrupted Laplacian, diffusion, and RFF features as capacity controls. Even after selecting the strongest of zero, shuffle, noise, and temporal-shift variants, corrupted features outperform z -only in only 2/81 dataset–encoder–feature settings. So, additional feature dimensions or fusion capacity alone are insufficient to explain the forecasting gains.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009

Table 10: Exploratory 3D video TDA results. Frames are treated as a spatiotemporal volume and cubical persistence features are appended for prediction. Each row reports a single run, comparing the no-TDA baseline against the best 3D-TDA mode by weighted MSE. Lower MSE is better.

Dataset	No TDA	Best 3D-TDA	Best mode	Δ	Pixel MSE
Bouncing rings	0.1432	0.1323	$H_0 + H_1$	+7.6%	0.0796
Bouncing disks	0.1509	0.1402	$H_0 + H_1$	+7.1%	0.0711
Orbiting rings	0.1506	0.1488	$H_0 + H_2$	+1.2%	0.0677
Orbiting disks	0.1765	0.1751	H_0	+0.8%	0.0971
Moving MNIST	0.2269	0.2264	$H_0 + H_1 + H_2$	+0.2%	0.0987
Glioblastoma	0.0220	0.0217	H_0	+1.4%	0.0020
HeLa	0.1410	0.1202	$H_0 + H_2$	+14.8%	0.0112
Lorenz-96	0.0233	0.0227	H_2	+2.6%	0.0130
ElectricDevices	0.0477	0.0474	$H_0 + H_1 + H_2$	+0.6%	0.0304

Table 11: Single-run frozen V-JEPA robustness check. V-JEPA is frozen and only the temporal predictor/fusion head is trained. We cap the run at 64 training sequences and 32 test sequences. Lower MSE is better; Gain is the relative MSE reduction of the best topology-fusion mode over raw V-JEPA z.

Dataset	V-JEPA z MSE	Best topo mode	Best topo MSE	Best topo R^2	Gain
Bouncing rings	0.0228	H1	0.0220	0.9780	3.5%
Bouncing disks	0.0215	PI-H1	0.0193	0.9806	10.2%
Orbiting rings	0.0294	PersLay-H1	0.0272	0.9726	7.5%
Orbiting disks	0.0301	PI-H1	0.0288	0.9711	4.3%
Moving MNIST	0.0394	PersLay-H1	0.0371	0.9629	5.8%
Glioblastoma	0.0290	PersLay-H1	0.0289	0.9708	0.3%
HeLa	0.0632	PersLay-H1	0.0629	0.9358	0.5%
Lorenz-96	0.0116	PI-H1	0.0095	0.9905	18.1%
ElectricDevices	0.0430	PersLay-H1	0.0480	0.9526	-11.6%

Persistent Latent Trajectory Features for Sequential Forecasting

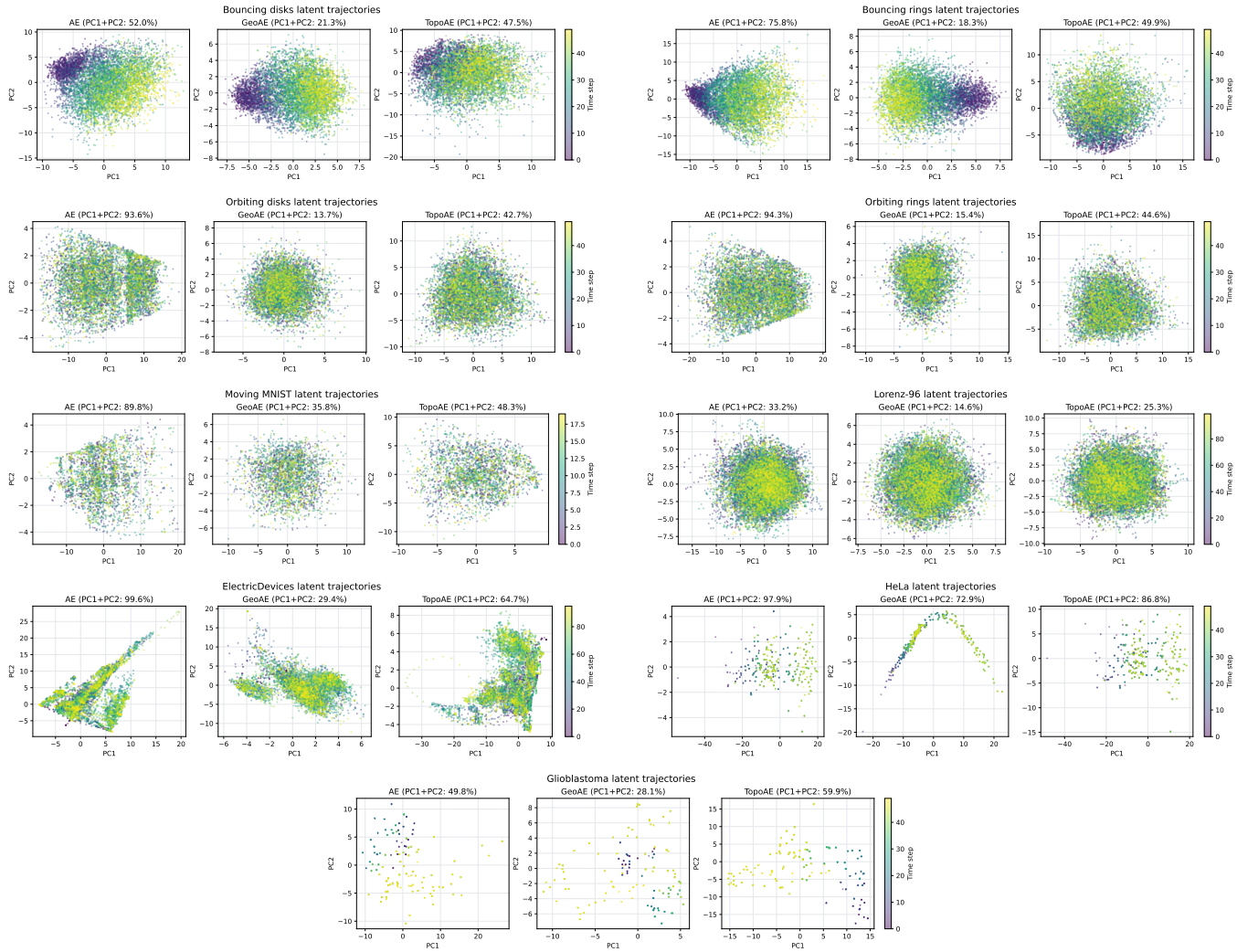


Figure 6: PCA projections for each dataset and encoder variant. Each panel header reports the percentage of total latent variance explained jointly by PC1 and PC2.

Table 12: Ablation over fused persistence-diagram vectorizations for the standard, geometry-regularized, and topology-regularized autoencoders. Bold marks the best vectorization within each dataset–encoder row. Lower MSE is better.

Dataset	Encoder	Betti H_1	PI H_1	Landscape H_1	PersLay-style H_1
Bouncing disks	AE	0.3639±0.0297	0.3656±0.0293	0.3682±0.0296	0.3671±0.0259
	GeoAE	0.0025±0.0003	0.0026±0.0003	0.0026±0.0003	0.0019±0.0007
	TopoAE	0.2259±0.0075	0.2263±0.0120	0.2356±0.0108	0.2293±0.0081
Bouncing rings	AE	0.3112±0.0561	0.3162±0.0546	0.3189±0.0522	0.3189±0.0573
	GeoAE	0.0023±0.0002	0.0022±0.0002	0.0027±0.0004	0.0023±0.0002
	TopoAE	0.2160±0.0105	0.2168±0.0126	0.2255±0.0122	0.2175±0.0113
Orbiting disks	AE	0.0070±0.0028	0.0069±0.0029	0.0068±0.0027	0.0069±0.0027
	GeoAE	0.0015±0.0001	0.0016±0.0001	0.0015±0.0001	0.0017±0.0001
	TopoAE	0.2761±0.0162	0.2857±0.0172	0.2851±0.0132	0.2869±0.0210
Orbiting rings	AE	0.0056±0.0019	0.0057±0.0020	0.0055±0.0021	0.0054±0.0020
	GeoAE	0.0017±0.0001	0.0017±0.0001	0.0017±0.0001	0.0016±0.0002
	TopoAE	0.2617±0.0202	0.2712±0.0305	0.2796±0.0298	0.2714±0.0253
Moving MNIST	AE	0.0087±0.0028	0.0085±0.0033	0.0080±0.0030	0.0081±0.0031
	GeoAE	0.0027±0.0005	0.0027±0.0004	0.0026±0.0005	0.0023±0.0005
	TopoAE	0.3037±0.0119	0.3057±0.0131	0.3126±0.0146	0.3103±0.0117
Lorenz-96	AE	0.1998±0.0223	0.2058±0.0275	0.2039±0.0226	0.2061±0.0244
	GeoAE	0.0017±0.0005	0.0013±0.0003	0.0016±0.0003	0.0019±0.0009
	TopoAE	0.3264±0.0258	0.3446±0.0175	0.3527±0.0158	0.3318±0.0212
ElectricDevices	AE	0.0167±0.0094	0.0171±0.0086	0.0175±0.0100	0.0168±0.0092
	GeoAE	0.0017±0.0001	0.0019±0.0003	0.0022±0.0002	0.0017±0.0002
	TopoAE	0.1210±0.0144	0.1120±0.0063	0.1415±0.0114	0.1401±0.0165
HeLa	AE	0.0496±0.0081	0.0579±0.0052	0.0537±0.0088	0.0507±0.0089
	GeoAE	0.0146±0.0033	0.0255±0.0100	0.0153±0.0056	0.0133±0.0054
	TopoAE	0.0644±0.0351	0.0881±0.0225	0.0685±0.0402	0.0676±0.0387
Glioblastoma	AE	0.0100±0.0024	0.0104±0.0051	0.0107±0.0033	0.0071±0.0016
	GeoAE	0.0063±0.0035	0.0209±0.0217	0.0076±0.0028	0.0075±0.0021
	TopoAE	0.0078±0.0015	0.0100±0.0016	0.0089±0.0021	0.0096±0.0034

Table 13: Compact summary of the full tuned sweep. Entries are mean MSE; the complete mode-level mean $\pm$ SD results are retained in the supplementary raw summary. “Topology only” excludes z , while direct and fusion combine topology with z . “Control” is the best PCA, KPCA, Laplacian, diffusion, or RFF variant. Bold marks the best entry in each dataset–encoder row. Lower is better.

Dataset	Encoder	z	Topology only	Best direct	Best fusion	Best control
Bouncing disks	AE	0.3429	0.5289	0.3402	0.3639	0.3437
	GeoAE	0.0013	0.0082	0.0010	0.0019	0.0014
	TopoAE	0.2158	0.2943	0.2070	0.2259	0.2152
Bouncing rings	AE	0.2972	0.4238	0.2955	0.3112	0.2937
	GeoAE	0.0013	0.0080	0.0011	0.0022	0.0013
	TopoAE	0.2104	0.2701	0.2024	0.2160	0.2082
Orbiting disks	AE	0.0052	0.0190	0.0047	0.0068	0.0051
	GeoAE	0.0017	0.0105	0.0011	0.0015	0.0016
	TopoAE	0.2281	0.3611	0.2083	0.2761	0.2235
Orbiting rings	AE	0.0045	0.0169	0.0040	0.0054	0.0044
	GeoAE	0.0017	0.0109	0.0010	0.0016	0.0014
	TopoAE	0.2321	0.3221	0.2254	0.2617	0.2326
Moving MNIST	AE	0.0048	0.0336	0.0046	0.0080	0.0046
	GeoAE	0.0009	0.0197	0.0009	0.0023	0.0009
	TopoAE	0.2868	0.4002	0.2748	0.3037	0.2895
Lorenz-96	AE	0.1536	0.2106	0.1420	0.1998	0.1441
	GeoAE	0.0012	0.0058	0.0012	0.0013	0.0012
	TopoAE	0.2726	0.5428	0.2745	0.3264	0.2726
ElectricDevices	AE	0.0139	0.0927	0.0102	0.0167	0.0118
	GeoAE	0.0014	0.0081	0.0011	0.0017	0.0014
	TopoAE	0.0615	0.2020	0.0492	0.1120	0.0604
HeLa	AE	0.0437	0.0433	0.0383	0.0496	0.0409
	GeoAE	0.0072	0.0300	0.0071	0.0133	0.0079
	TopoAE	0.0587	0.0619	0.0550	0.0644	0.0558
Glioblastoma	AE	0.0027	0.0100	0.0021	0.0071	0.0022
	GeoAE	0.0011	0.0730	0.0008	0.0063	0.0012
	TopoAE	0.0040	0.2179	0.0033	0.0078	0.0037